Proof Spine for *Distinction and Continuation* and *The Shape of Possibility*

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Each load-bearing result of the two volumes appears below with its status. **E** marks an established result, to be attributed to its source; **N** marks a new result to be proved in full under the listed assumptions; **O** marks a proof obligation. A result with status O may not be cited as a theorem by any later chapter. Volume 1 is *Distinction and Continuation* and Volume 2 is *The Shape of Possibility*.

ID	Statement	St.	Where	Assumptions / source	Depends on
S1	A decision factors through a representation iff equal representations force equal decisions.	E	V1 Ch. 2; both Ch. 1	Standard quotient factorization.	none
S2	If $\sigma(h_1) = \sigma(h_2)$ and $P(h_1) \neq P(h_2)$, no procedure receiving σ alone is correct on both histories.	E	both Ch. 1	Corollary of S1.	S1
S3	Under a sound gate that consults σ alone, a continuation admissible at h_1 is refused whenever some h_2 with the same representation is inadmissible.	N	both Ch. 1	Soundness over all of $\mathcal{H}$; gate depends on $\sigma(h)$ only. Elementary.	S1
S4	Representational obstruction can be inherited from an obligation obstruction at a different history.	N	both Ch. 1	Example, not a theorem.	S3
S5	Coarsest representation deciding a fixed policy is the kernel of the policy.	E	V1 Ch. 5	Standard.	S1
S6	For a fully separating policy family, every sufficient representation is injective.	N	V1 Ch. 6	Separating means distinct histories are distinguished by some policy in the family. Elementary.	S5
S7	Minimal automaton and continuation equivalence (Myhill–Nerode).	E	V1 Ch. 7	Nerode (1958); Myhill (1957).	none
S8	Gap between one-shot and update-maintainable sufficiency, characterized by the Myhill–Nerode congruence over the update family.	N	V1 Ch. 7	s_{one} finite for the equality criterion; counter-families P_n and q_{neg} given. Proved in full.	S7
S9	Tight state complexity mn for the intersection of two independent modular counters over disjoint alphabets.	E	V1 Ch. 8	Yu, Zhuang, Salomaa (1994); generalized in S36.	S7
S10	Admissibility decomposes into physical executability, authorization, and obligation satisfaction; the normative obstruction has four forms, one of them joint.	N	V1 Ch. 13	Prefix-closed sets; definitional plus elementary.	none

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S11	Monitor for bounded-response obligations over disjoint alphabets: the minimal monitor has exactly $\prod_i (k_i + 1) + 1$ states.	N	V1 Ch. 14	Fragment fixed (Definition of the disjoint fragment), counting in each component's own events. Shared-clock case remains open. Proved in full.	S7, S36
S12	Coarsest identity criterion preserving promised observations is their joint kernel; preserving a promise with updates (a submonoid) it is the continuation equivalence of the family.	N	V1 Ch. 15	Elementary; F a submonoid. Proved in full.	S5, S35
S13	Admissible shedding characterized from dependency structure (disjoint from a support); no additive capacity measure suffices when alternatives exist.	N	V1 Ch. 17	Finite Horn dependency system and required set given; proved in full (see S42).	S10
S14	Coarsening a representation contracts the family of implementable policies; later computation cannot recover discarded distinctions.	N	V1 Ch. 21	$\sigma' = f \circ \sigma$. Elementary, via S1.	S1
S15	Belief state is a sufficient statistic in a POMDP.	E	V2 Ch. 5	Åström; Smallwood and Sondik.	none
S16	Belief state is minimal sufficient for a stated policy family.	O	V2 Ch. 5	Policy family and observation model to be fixed.	S15, S5
S17	Augmented state sufficient for feasibility iff the added component is sufficient in the V1 sense.	N	V2 Ch. 7	Feasibility is a function of the history. Elementary.	S5
S18	Discrete-time viability kernel is the greatest fixed point of the one-step predecessor operator.	E	V2 Ch. 3	Aubin; standard.	none
S19	Reachability decidable for Petri nets; coverability EXPSPACE-hard.	E	V2 Ch. 6	Mayr; Kosaraju; Lipton.	none
S20	Viability kernels for systems whose constraint set is updated by an obligation monitor.	N	V2 Ch. 14	Bounded-response monitor of S11 in product form; kernel monotone in slack; corridor example in full.	S11, S18
S21	Invariant preservation for explicit/automatic switching under dwell time and monitoring latency.	N	V2 Ch. 17	Each controller keeps the safe set invariant alone; latency bound; failure model. Liberzon; Hespanha and Morse.	S18
S22	Interface preserves admissible reachability iff a restricted simulation relation exists.	N	V2 Ch. 18	Admissible region fixed; abstraction total on it.	S10
S23	Invariance of a constraint set under rule changes restricted to an explicit relation, including constraints on revising constraints.	N	V2 Ch. 19	Rule-change relation and its own constraint stated.	S18
S24	Proxy-optimal policy is $2\varepsilon/(1-\gamma)$-optimal under uniform reward error ε.	E	V2 Ch. 20	Discounted MDP; Puterman.	none
S25	Constrained-MDP version of S24 with proxied objective and constraints.	N	V2 Ch. 20	Uniform bound on both objective and constraint error; feasibility may fail.	S24
S26	Cost bounds for maintaining a representation sufficient for an anticipated policy family.	N	V2 Ch. 22	Cost model to be fixed.	S14, S6

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S27	Continuation quotient of the account: classes are $\perp$ (overdrawn) and the balances $n \in \mathbb{N}$ otherwise.	N	V1 Ch. 2	Positive integer amounts; overdraft permitted; all finite event sequences. Proved in full.	S1
S28	Under a size bound $2 \leq k < n$ there is no most informative representation of an n-element set, and at least two incomparable maximal ones.	N	V1 Ch. 3	X finite; size is number of values. Proved in full.	S1
S29	Myhill–Nerode construction for properties of histories: continuation equivalence is the coarsest right congruence inside $\ker P$, and the Nerode system is minimal.	E	V1 Ch. 4	Myhill; Nerode.	S1
S30	Every behavioural property is the intersection of a safety and a liveness property.	E	V1 Ch. 4	Alpern and Schneider (1985).	none
S31	Nonempty safety properties correspond bijectively to pruned history sets via the viable histories.	E	V1 Ch. 4	Closed sets and pruned trees; Kechris.	S30
S32	On viable histories of the no-overdraft property, the set of admissible extensions determines the balance.	N	V1 Ch. 4	Positive integer amounts. Proved in full.	S27, S31
S33	Sufficiency for a fixed policy: $\ker r \subseteq \ker \pi$; no non-injective representation suffices for all two-action policies; minimal representations are sensitive to a one-case change; sufficiency depends on the domain.	E, N	V1 Ch. 5	Theorem E; the other three N, elementary, proved in full.	S1
S34	Joint sufficiency for policy families; separating families force injectivity; threshold chains cost $K + 1$ values.	E, N	V1 Ch. 6	Bounds and VC bound E (Sauer–Shelah, Assouad); separating and chain results N. Sharp bounds remain a research problem.	S5, S6
S35	Maintainable and F-maintainable sufficiency; Nerode theorem for update families; minimal machine by Moore partition refinement.	E, N	V1 Ch. 7	Moore (1956), Hopcroft (1971) cited; F-congruence theorem N.	S7, S8
S36	Independent properties over disjoint alphabets: minimal memory is the product of the classes; sufficiency composes but minimality does not.	N	V1 Ch. 8	Alphabet partition with each property depending on its own part. Proved in full.	S7, S9
S37	Exact sufficiency is the only preservation holding under every distribution; Bayes error is monotone under coarsening and errors add across policies (tight); zero conditional entropy characterizes sufficiency.	E, N	V1 Ch. 9	Fano and data-processing inequalities E. Others N, finite X, proved in full.	S5
S38	Verified region of a claim is its interior in the partition induced by the evidence; a proof reaches a case only through evidence settling its assumptions.	E, N	V1 Ch. 10	Certifying algorithms E (McConnell et al.; Necula). Interior and gap results N, proved in full.	S1
S39	Hash chain determines the sequence (idealized injective hash); vector clocks; provenance semirings. Minimal last-author record has $(\lvert U\rvert + 1)^K$ values; content identity does not determine lineage.	E, N	V1 Ch. 11	First three E (Merkle; Fidge, Mattern; Green et al.). Counts and separation N, proved in full.	S6, S36

ID	Statement	St.	Where	Assumptions / source	Depends on
S40	Parity needs n probes and a marked cell $n-1$; derived cells leave storage adequacy unchanged while collapsing retrieval depth.	N	V1 Ch. 12	Deterministic cell-probe readers; worst case over the domain. Proved in full.	S1
S41	Exactly-once effects under identity, atomic record, and fair channels; non-atomic variants give at-most-once or at-least-once; deduplication memory is exactly $2^{\lvert I\rvert}$ states under unbounded duplication.	E, N	V1 Ch. 16	Two generals impossibility E (Akkoyunlu et al.; Gray; Halpern and Moses). Rest N, proved in full.	S7
S42	Admissible shed sets are those disjoint from a self-supporting set; scalar capacity tests are correct for conjunctive dependencies and fail with alternatives (counterexample). Smallest support is set-cover hard.	N	V1 Ch. 17	Finite Horn dependency systems. Proved in full; hardness by standard reduction.	S10, S11
S43	Restoration and replacement defined relative to an identity criterion; no compensation restores a step crossing a bad prefix; aborted sagas restore iff criterion is a congruence for compensations (counterexample otherwise).	N	V1 Ch. 18	Prefix-closed obligation sets; per-step restoration. Proved in full.	S12, S30
S44	Global property decidable from causal past iff insensitive to concurrent events; quota theorem: independent capacities of replicas sum to at most the common balance.	N	V1 Ch. 19	Finite executions, integer withdrawals. CRDT convergence and invariant confluence E (Shapiro et al.; Bailis et al.). Proved in full.	S39
S45	Refinement preserves compatibility of deterministic interface automata; non-circular assume–guarantee rule under alphabet conditions; attenuating delegation creates no authority.	N	V1 Ch. 20	Deterministic automata; prefix-closed components. Nondeterministic version E (de Alfaro, Henzinger). Proved in full.	S10
S46	Revision horizon determines the representation and shrinks by $\lvert A\rvert^{-(k-k')}$ under coarsening; affected cases are the non-constant classes; worst-case binary revision error under the uniform distribution is $\frac{1}{n}\sum_y \lfloor n_y/2 \rfloor \geq \frac{n-k}{2n}$, against $\approx 1/2\varepsilon$ values for thresholds.	N	V1 Ch. 21	Finite X, uniform μ. Proved in full; general approximate revision sufficiency open.	S1, S14, S37
S47	Representation selection is weighted set cover over disagreement pairs: greedy $H(\lvert D\rvert)$ approximation, NP-hard (injective policy is minimum test set), checking is $O(\lvert X\rvert\lvert F\rvert)$; updatable storage for P_n costs exactly $n+1$ values.	E, N	V1 Ch. 22	Minimum test set and greedy cover E (Garey–Johnson; Chvátal). Rest N, proved in full.	S8, S35
S48	Worked systems: 13-state monitor and gate optimality; document record adequacy; checkout service supports; ledger semantics.	N	V1 Ch. 23	Each instantiates S11, S13, S39. Proved in full.	S11, S39, S42
S49	SCC partition is the minimal sufficient representation for the reachability family; irreversible edges are exactly the inter-component edges; at most $c-1$ component changes.	N	V2 Ch. 2	Finite transition graph. Tarjan cited. Proved in full.	S1

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S50	Greatest fixed point of the predecessor operator is the viability kernel and the set of states with infinite K-trajectories; viable histories are those ending in the kernel; kernel convex for linear systems with convex U, K.	N	V2 Ch. 3	Finite branching or compactness as stated. Aubin cited.	S18
S51	Almost-sure safety is the support-based viability kernel; optimizing under it reduces to the restricted MDP; expected-cost constraints can force randomization.	N	V2 Ch. 4	Finite MDP. Bellman contraction and occupation measures cited.	S18
S52	Predictive equivalence of beliefs is the orthogonal complement of the span of prediction vectors; belief minimal iff rank condition; almost-sure safety decided on supports.	N	V2 Ch. 5	Finite POMDP. Proved in full; refines S15.	S15
S53	State equation necessary not sufficient for Petri net reachability; place invariants bound markings; the viability set of a net is upward closed with finite basis.	N	V2 Ch. 6	Nonnegative integer weights. Dickson cited.	S19
S54	Augmentation theorem: least augmentation has size equal to the fibre count; hard budgets are exact by the remaining-budget product; unbounded counters admit no finite augmentation.	N	V2 Ch. 7	Maintainable state description. Completes S17.	S17, S29
S55	Return cost is a quasi-metric whose finite part is reversibility; budget above $(\|X\|-1)c_{\max}$ buys nothing; path dependence is insufficiency of the state; policies refine components.	N	V2 Ch. 8	Finite graph, nonnegative costs. Norris cited for absorption.	S49, S54
S56	In monotone capability systems reachable configurations have a unique maximum, the closure, computed in linear time; shortest acquisition plans have length at most $\|\mathrm{Cap}\setminus C\|$; nominal, eventual, current menus are nested.	N	V2 Ch. 9	Monotone (no losses). Bylander cited for the non-monotone case.	S49
S57	Acquisition with costly inspection: backward induction on a finite sufficient state, oracle bound with explicit gap, submodularity of domain value, budget-admissible optimum.	N	V2 Ch. 10	Independent finite-support values for the DP; submodularity needs only nonnegativity. Weitzman cited.	S54
S58	Cardinality, volume, and graph distance of reach misrank budgeted freedom; dominance is a preorder with incomparable pairs and weighted scores reverse them; simulation preserves coverage; growth rate of feasible histories is the largest spectral radius among reachable components.	N	V2 Ch. 11	Finite graph. Milner; Perron–Frobenius cited.	S49

ID	Statement	St.	Where	Assumptions / source	Depends on
S59	Cost algebras change the optimal path; componentwise optima not jointly achievable; scalar weights decide convex mixtures; Pareto front can have 2^n points; integer vector budgets decided by product; shared acquisition subadditive; time and energy of task graphs.	N	V2 Ch. 12	Finite graphs, nonnegative costs. Lehmann, Garey–Johnson, Graham cited.	S55, S56
S60	Robust kernel is the greatest fixed point for arbitrary spaces; monotone in disturbance set; sharp threshold $\delta = \frac{1}{2}$ in the scalar example; Gaussian chance-constraint equivalent; Boole allocation; no forever-safety under unbounded noise; constant parameters can be strictly easier than varying ones.	N	V2 Ch. 13	History-dependent policies; bounded or full-support noise as stated.	S18, S51
S61	Compliant viability is the kernel of the product with the obligation monitor; monotone in slack; a trigger is admissible iff its successor is in the kernel; explicit corridor kernel; trigger admissibility depends on pending obligations; no sound and complete monitor-only gate.	N	V2 Ch. 14	Finite deterministic system, disjoint trigger and response sets. Completes S20.	S11, S18
S62	Viable optimization is optimization on the kernel; price of viability zero iff an unconstrained optimum stays in K and unbounded in general; lookahead $\lvert X\rvert$ needed and sufficient; m-option kernels nested, branching m^k, prices nondecreasing in m.	N	V2 Ch. 15	Finite deterministic discounted system.	S18, S51
S63	Recoverable region is the least fixed point of the backward step and predecessor closed; progress functions bound recovery time by φ and greedy descent with any fixed lookahead can leave the region; attractor strictly smaller than possible recovery; fallback regions of shedding are down-closed, linear for a unique support, NP-complete in general, exponentially many maximal elements.	N	V2 Ch. 16	Finite systems; dependency systems of S42.	S42, S49
S64	Hybrid switching: a common invariant set needs no switching conditions; distinct invariant sets do not compose; with a common core, handover time and latency the safe set is preserved under a dwell-time clause; sharp example; tolerable latency equals exit distance.	N	V2 Ch. 17	Finite or compact systems; fault model for the automatic controller.	S21
S65	Interfaces as macro denotations: admissible reach is characterized by weak simulation in both directions (completeness and soundness); one-step sufficiency is not necessary; forced reachability of the presented system lies in possible recovery of the concrete one.	N	V2 Ch. 18	Admissible step relations fixed.	S22

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S66	Augmented rule systems: the robust kernel has sections that are viability domains relative to the intersection of amendment sections; the amendment depth is a shortest path in the revision graph; unrestricted revision gives depth one; every depth is realized.	N	V2 Ch. 19	Finite rule and state sets; amender acts after the controller.	S23, S60
S67	Proxy transfer is tight at $2\varepsilon/(1-\gamma)$; on-policy refinement charges error on the two occupancy measures; constrained version with tightened budget has loss 2ρ plus a concavity-bounded price; hard safety on a viability domain is independent of the proxy.	N	V2 Ch. 20	Finite MDPs; discounted criterion.	S24, S25
S68	Untried versus infeasible: trichotomy of attempted, certified infeasible and untested; data on attempted alternatives are silent on independent untested ones; threshold horizon $n^* = 1 + c/(p\Delta)$ and minimax regret; knowledge-indexed viability kernel is monotone in knowledge and safe attempts are those with all outcomes inside it.	N	V2 Ch. 21	Static feasibility; revealed outcomes deterministic.	S18, S12
S69	Least maintainable representation sufficient for a family is the intersection of Nerode congruences, with between the largest and the product of class counts; marginal cost of one more property; rent-or-buy with ratio 2 optimal among deterministic rules; irreversible loss forces indefinite maintenance; shared rent.	N	V2 Ch. 22	Cost model of Def. 22.2; exponential demand for the stochastic parts.	S26, S14
S70	Flexibility-constrained design: commit iff $g > hR$ independently of horizon; polynomial for single-route interventions, NP-complete with two routes (vertex cover); flexibility price is the gain of the union of routes and decreases under refinement.	N	V2 Ch. 23	Additive gains; independent exponential demand.	S66, S69
S71	Four case studies: vehicle kernel $b \geq x$ and robust $b \geq (1+\delta)x$; archive amendment depth and 8-value representation; adaptive controller kernels K, $\{-3..1\}$, $\varnothing$; network fallback region, redundancy is not capacity.	N	V2 Ch. 24	Finite instances computed exactly.	S66, S69, S67
S72	Appendices B to G (self-contained recaps): walk counts grow at the spectral radius; linear-time kernel algorithm with counters; scalar affine invariant intervals; Lyapunov recovery time; concavity of LP value in the budget; value-iteration stopping rule; memoryless laws; Monte Carlo sample size.	N	V2 App. B–G	Standard background with short proofs; cited results marked established.	S18, S58

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S73	Book 1 appendices A to F: induction on histories, finite Knaster–Tarski, fooling sets, adversary and exchange arguments; representations form a partition poset with products and coproducts, quotients as coequalizers, Galois connection of questions and representations; Moore minimization and $(k+2)$-state bounded-response monitor; data processing and sufficiency, non-recovery bound; self-contained recap of reachability and the viability kernel with a linear-time algorithm.	N	V1 App. A–F	Finite sets throughout; classical results marked established.	S18, S29
S74	Expansion of Chapters 1 to 10: representation without the gate bit permits no sample; reach profile stabilizes in at most $n-1$ steps; DAG of components and path counts; kernel algebra and stopping curve; geometric ruin and one-random-state bound; down-closed winning supports; backward coverability and Karp–Miller; decidability of sufficiency and normal-form rate-limit states; Eswaran–Tarjan augmentation and expected return cost; depth versus plan length and non-submodular closure; spread raises the reservation value (verified instance $V^* = 8.4$, oracle 9.6).	N	V2 Ch. 1–10	Finite systems; instances computed by script.	S10, S18, S31
S75	Dominance is the componentwise order on cost vectors; simulation is monotone for label growth; rate-limit growth rates φ and 1.4656 from Fibonacci recurrences.	N	V2 Ch. 11	Finite graphs; deterministic for $L_k = N_k$.	S40
S76	Lagrangian bound equals the mixture optimum (gap 6 against 4 on a three-path mission); list-scheduling anomaly (12 with three workers, 15 with four).	N	V2 Ch. 12	Finite path set; Graham list rule.	S41
S77	Distribution-free chance margin $\sqrt{(1-\varepsilon)/\varepsilon}$ against Gaussian margin; horizon allocation numbers; asymptotic geometric exit time via Krein–Rutman, with survival table.	N/E	V2 Ch. 13	Strictly positive Gaussian kernel on a compact set.	S42, S43
S77a	Exact admissibility of a second trigger in the corridor (two orderings); corridor kernel count $(L+1)+\sum_{c\le k}\min(c, L+1)$.	N	V2 Ch. 14	Responses at the two ends; verified on 2646 cases.	S44
S78	The unit-control speed-limited line has $W_2 = \varnothing$; doubling authority gives kernels of sizes $77, 59, 0$; fault tolerance is a transversal number minus one (navigation stack, $\lvert\mathrm{FB}(3)\rvert = 20$, $\lvert\mathrm{FB}(4)\rvert = 31$).	N	V2 Ch. 15–16	Finite systems; counts by exhaustive computation.	S45, S46

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S79	Exact safe region $\{\ell \le d,\ \tau \ge \ell\}$ for relegation in the sharp family; minimum interface and size–effort trade-off; depth of a quorum rule equals the sum of the q cheapest capture costs.	N	V2 Ch. 17–19	Families as stated; game solved exhaustively for four parameter pairs.	S47, S48, S49
S80	Selection on a noisy Gaussian proxy delivers $m_n/\sqrt{1+\sigma^2}$ and the proxy overstates by exactly $1+\sigma^2$.	N	V2 Ch. 20	Independent Gaussian values and noise.	S50
S81	Threshold for attempting once in a Bayesian two-armed instance, and the exact optimal values by dynamic programming; counters on one event need lcm classes; randomized rent-or-buy ratio 1.5353 at $B = 10$.	N/E	V2 Ch. 21–22	Uniform prior; one event; discrete rent-or-buy.	S51, S52
S82	Flexibility price equals minimum weight vertex cover (weighted instance: loss 14 optimal, matching heuristic 29, local ratio 16); chance kernel of the vehicle is $b \ge x +$ Bin-quantile.	N/E	V2 Ch. 23–24	Single-module routes; independent two-valued costs.	S53, S54